International Trade[1]

Lecture Note 2: Ricardo with a Continuum of Goods and Two Countries

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[1]The notes are still preliminary. If you find any typo or mistake, please send it to malfaro@ualberta.ca.

Contents

1 Ricardian Models **1**

2 Setup **1**

2.1 Supply . . . 2

2.2 Demand . . . 3

3 Autarky **3**

3.1 Autarky Equilibrium . . . 4

3.2 Welfare in Autarky . . . 5

4 Free Trade **6**

4.1 Wages and Goods Prices . . . 6

4.2 Demand side . . . 7

4.3 Balanced Trade . . . 8

4.4 Equilibrium . . . 9

4.5 Gains of Trade . . . 9

4.6 The Role of Absolute Advantages . . . 10

5 Free Trade: Differences in Market Size **11**

5.1 Results . . . 11

5.2 Interpretation . . . 12

5.3 Impact on Welfare . . . 13

6 Free Trade: Improvements in Technology of the Trading Partner **14**

6.1 Result . . . 14

6.2 Increases in H's Welfare . . . 16

6.3 Decreases in H's Welfare . . . 17

7 Costly Trade and Nontradable goods **18**

Notation

This is a derivation

This is some comment

This is a comment on advanced topics which are not part of the course (you can ignore it without loss of continuity regarding the text)

- The symbol ":=" means "by definition".
- Vectors are denoted by bold lowercase letters (for instance, $\mathbf{x}$) and matrices by bold capital letters (for instance, $\mathbf{X}$).
- To differentiate between the verb "maximize" and the operator "maximum", I denote the former with "max" and the latter with "sup" (i.e., supremum). The same notation applies to "minimize" and "minimum", where I respectively use "min" and "inf" (the latter indicates infimum).
- "iff" means "if and only if"
- $\exp(x)$ is the function e^x.
- Random variables are denoted with a bar below. For instance, $\underline{x}$.

These notes contain hyperlinks in blue and red text. If you are using Adobe Acrobat Reader, you can click on the link and easily navigate back by pressing Alt+Left Arrow.

1 Ricardian Models

These notes introduce one of the foundational models in International Trade: the Ricardian model. In particular, we present the version built by **Dornbusch, Fischer, and Samuelson (1977)** (henceforth, DFS).

Unlike Ricardo's original formulation, which considers two goods and two countries, DFS introduces **a continuum of goods in a two-country setting**. This framework conceptualizes each good as a point on an interval, enabling the use of calculus techniques and other simplifications. As a result, the model becomes highly tractable, particularly compared to models with a discrete number of goods.

The goal of studying DFS is twofold. First, it helps us become familiar with techniques employed in frameworks featuring a continuum of goods. Second, it unveils the mechanisms and main insights behind the Ricardian model.

Note that DFS is somewhat limited for empirical applications, due to its assumption of only two countries. Despite this limitation, a deep understanding of DFS is still crucial as, ultimately, similar mechanisms are at play in all Ricardian models.

In the next lecture, we will discuss a generalization of the Ricardian model by Eaton and Kortum (2002). This has become a seminal work over the past two decades. It overcomes the two-country limitation, by considering a continuum of goods and multiple countries. Furthermore, it proposes a set of functional forms to take the model to the data.

2 Setup

There is a world economy comprising two countries. We respectively refer to them as Home (H) and Foreign (F). To differentiate variables belonging to one country from the other, any variable belonging to (F) includes a tilde.

The economy consists of a continuum of industries, with each comprising a single homogeneous good. Formally, the set of industries is $J := [0, 1]$, with a typical element denoted $j \in J$. Since each industry produces a single good, we will employ the terms *goods* and *industries* interchangeably.

Moreover, countries have a fixed number of agents, where L denotes the number of agents in (H) and $\widetilde{L}$ in (F). In both countries, agents make consumption decisions and supply labor, working for the industry that offers the highest wage. To simplify the analysis, we assume that each agent supplies one unit of labor inelastically, implying that labor supply is independent of the prevailing wage. While labor is perfectly mobile across industries within a country, it is immobile across countries.

2.1 Supply

The market structure is characterized by *perfect competition*. Furthermore, each good is produced using labor as the sole input, with all workers exhibiting the same level of productivity within a country.[1]

All firms within a country have access to the same producing technology, with technologies differing across countries. Technology can be described by the **unit labor requirement** of good j, which represents the amount of labor necessary to produce one unit of good j. We respectively denote them by $a(j)$ and $\widetilde{a}(j)$ for each country. Since labor is the sole input, unit labor requirements completely identify the **marginal productivity of labor**. This corresponds to the inverse of the unit labor requirements: $\frac{1}{a(j)}$ for (H) and $\frac{1}{\widetilde{a}(j)}$ for (F).

We assume that unit labor requirements are constant. This implies that production exhibits constant returns to scale, which in turn ensures constant marginal costs.

We define the relative efficiency of good j by $A(j) := \frac{\widetilde{a}(j)}{a(j)}$. Without loss of generality, we order the set of goods J according to a decreasing order of home-country relative efficiency, so that $A'(j) < 0$.[2] The order is such that (H) has comparative advantages in low-indexed goods $j \in J$, while (F) has comparative advantages in high-indexed goods $j \in J$.

As we will see, the patterns of production are determined by the concept of **comparative advantages**, which are based on relative efficiencies. Absolute efficiencies $a(j)$ and $\widetilde{a}(j)$ on their own are not sufficient to identify which goods each country produces.

[1] All the conclusions we obtain hold if the input requirements can be described through a single number. This requires that the production function exhibits constant returns to scale.

[2] Establishing this order is possible due to the existence of a single input. Another scenario in which an order can be defined is when aggregate multiple inputs can be aggregated into a single composite factor. Ultimately, the existence of an order requires that efficiency can be represented through a real-valued number.

2.2 Demand

We assume that preferences in both countries are homogeneous. They are represented by a Cobb-Douglas utility function with identical parameters. Denoting $q(j)$ as the consumption of a typical household in (H), utility is

$$U\left[(q(j))_{j\in J}\right] := \exp\left[\int_0^1 \beta(j)\ln q(j)\ \mathrm{d}j\right], \tag{1}$$

where $\beta(j) > 0$ for any $j \in J$ and $\int_0^1 \beta(j)\ \mathrm{d}j = 1$. Notice that the βs do not have any script denoting the country, reflecting that preferences are identical across countries.

The budget constraint is given by

$$y = \int_0^1 p(j)\, q(j)\ \mathrm{d}j,$$

where y is the household's income. The maximization problem determines an optimal consumption of a good j given by:

$$q(j) = \beta(j)\frac{y}{p(j)},$$

where $\beta(j)$ represents the fraction of income spent on good j.[3] Note that **all goods are essential under a Cobb Douglas utility function**, meaning that each good is consumed in positive quantities: $q(j)$ can only equal zero if $p(j) \to \infty$.

The household's indirect utility function is

$$v(\mathbf{p}, y) := \frac{y}{\mathbb{P}},$$

with $\mathbb{P} := \exp\left[\int_0^1 \beta(j)\ln\left[\frac{p(j)}{\beta(j)}\right]\ \mathrm{d}j\right]$. Welfare will be evaluated directly through v, which represents **welfare per capita**. Given homotheticity, v can be interpreted as **real income**.

3 Autarky

We start the analysis by considering a country in autarky. The results obtained for a closed economy will serve as the baseline for comparing outcomes against free trade.

[3]Strictly speaking, β is the density function of the expenditure's shares, given a continuum of goods. In this context, some βs could exceed 1, although $\int_0^1 \beta(j)\ \mathrm{d}j = 1$. If you find it easier, just stick to an interpretation where there are a large number of discrete goods.

3.1 Autarky Equilibrium

In autarky, a country is in equilibrium when both the labor market and the goods market clear.[4] Let w represent wages and $p(j)$ denote the price of good j. Any equilibrium variable will be denoted with a superscript *aut*.

We start by identifying some features of the equilibrium. First, **all goods must be produced**. This follows because each good is essential from the consumer's point of view: at any finite price, there will be a positive consumption for every good. Since there is no possibility of obtaining goods from another country under autarky, all goods have to be produced domestically. This contrasts with the case of free trade, where a country could possibly specialize in a subset of certain goods, obtaining the rest from the other country.

Second, **wages paid in each industry must be equal**. This follows due to the free mobility of labor within each country, implying that workers will choose working in the industry with highest wages. In this context, if one industry paid more than the other, all workers would move to that industry. This would result in no production in any other industry, which contradicts that all goods must be produced in equilibrium.

Making use of these properties, let us now pin down the equilibrium price of each good j. Denote the wage prevailing in autarky by $w^{aut} := 1$, which is taken as the numéraire. Perfect competition determines that the price of good j equals its marginal cost:

$$p^{aut}(j) = w^{aut} a(j). \tag{2}$$

Additionally, given $y^{aut} = w^{aut}$, the consumption per capita of good j in autarky is given by $q^{aut}(j) = \beta(j) \frac{w^{aut}}{p^{aut}(j)}$. Total demand is $Q(j) := Lq(j)$.

Note that all the values on the right-hand side of (2) are known, so that the equation fully identifies the autarky price of each good j. Consequently, **prices and wages are determined solely by the supply side, and hence are independent of the quantities consumed**. Thus, any change in the demand side will only affect the quantities consumed, without any impact on prices or wages.

[4]In fact, it is the labor market plus a unitary mass of goods markets since the set of goods is $[0, 1]$.

3.2 Welfare in Autarky

We measure welfare per capita through the indirect utility function. In logarithms, this can be expressed by

$$\ln v^{aut}(\mathbf{p}, y) := \int_0^1 \beta(j) \ln\left(\frac{\beta(j)}{a(j)}\right) \mathrm{d}j.$$

Derivation. $v^{aut}(\mathbf{p}, y)$ is given by the utility function (1), evaluated at the consumption vector $\left(q^{aut}(j)\right)_{j\in[0,1]}$. Since $q^{aut}(j) = \beta(j) \frac{w^{aut}}{p^{aut}(j)}$, then

$$v^{aut}(\mathbf{p}, y) := \exp\left[\int_0^1 \beta(j) \ln\left[\beta(j) \frac{y^{aut}}{p^{aut}(j)}\right] \mathrm{d}j\right]$$

Applying logs to both sides and using that $y^{aut} = w^{aut}$ and $p^{aut}(j) = w^{aut}a(j)$:

$$\begin{aligned}\ln v^{aut}(\mathbf{p}, y) &= \int_0^1 \beta(j) \ln\left[\beta(j) \frac{w^{aut}}{w^{aut}a(j)}\right] \mathrm{d}j, \\ &= \int_0^1 \beta(j) \ln\left[\frac{\beta(j)}{a(j)}\right] \mathrm{d}j.\end{aligned}$$

The function v allows us to compare welfare before and after a change. In particular, it is easy by using the logarithmic version of it, so that welfare comparisons can be more easily determined by the logarithm of welfare ratios.

For instance, let's compare a situation where country (H)'s production technology changes. Specifically, (H) has unit labor requirements a in the baseline scenario, and then become $\widetilde{a}$. Suppose in particular that $a(j) > \widetilde{a}(j)$ for each j, representing a technological improvement. Then, using the logarithm of welfare ratios,

$$\ln \frac{v^{aut}(\mathbf{p}, y)}{\widetilde{v}^{aut}(\mathbf{p}, y)} = \int_0^1 \beta(j) \ln[A(j)] \mathrm{d}j.$$

Derivation. Taking logs and differences:

$$\ln v^{aut}(\mathbf{p}, y) - \ln \widetilde{v}^{aut}(\mathbf{p}, y) = \int_0^1 \beta(j) \ln\left[\frac{\beta(j)}{a(j)}\right] \mathrm{d}j - \int_0^1 \beta(j) \ln\left[\frac{\beta(j)}{\widetilde{a}(j)}\right] \mathrm{d}j$$

So that

$$\begin{aligned}\ln \frac{v^{aut}(\mathbf{p}, y)}{\widetilde{v}^{aut}(\mathbf{p}, y)} &= \int_0^1 \beta(j) \left[\ln \widetilde{a}(j) - \ln a(j)\right] \mathrm{d}j, \\ &= \int_0^1 \beta(j) \ln[A(j)] \mathrm{d}j.\end{aligned}$$

Since $A(j) := \frac{\widetilde{a}(j)}{a(j)} < 1$, we can conclude that $\int_0^1 \beta(j) \ln[A(j)] \mathrm{d}j < 0$. This implies $\ln \frac{v^{aut}(\mathbf{p},y)}{\widetilde{v}^{aut}(\mathbf{p},y)} < 0$ and therefore $v^{aut}(\mathbf{p}, y) > \widetilde{v}^{aut}(\mathbf{p}, y)$, so that technology enhancements are welfare-improving. Note that, since $\int_0^1 \beta(j) \ln[A(j)] \mathrm{d}j$ is a weighted average, the increase in welfare would be more pronounced if the improvements in technology correspond to goods whose βs are higher.

4 Free Trade

Free trade is characterized by the absence of any trade barriers. This is equivalent to assuming that all goods are perfectly tradable, entailing that there is no distinction between delivering goods domestically or internationally.

In open economies, equilibrium still requires that the goods and labor markets clear in each country, just as in the case of autarky. However, we also need to consider how each country's imports are financed. To address this, a standard assumption in the literature is to assume balanced trade, where the value of a country's exports equals the value of its imports[5]. An equivalent version of the same requirement is that each country's income equals its expenditure.

4.1 Wages and Goods Prices

Let's describe how prices are set in an open economy for given wages w and $\widetilde{w}$. First, notice that the price of good j in each country must be the same, entailing that **the law of one price must hold**. This is due to the total absence of trade costs, where importing a good entails no additional frictions or costs compared to domestic consumption.

Furthermore, since goods are homogeneous, consumers will buy the good j from the country with the lowest price. Jointly with the law of one price, this determines that the **country producing a good will be the sole supplier in the world**, thus selling it both domestically and abroad. This also implies that there cannot be simultaneous imports and exports of the same good.

Considering that prices are equal to marginal cost in each country, the price of j in any country will be given by the cheapest source:

$$p^{trade}(j) = \inf \{wa(j), \widetilde{w}\widetilde{a}(j)\}.$$

This equation has important implications for the production of goods. Specifically, any good $j \in J$ for which $wa(j) < \widetilde{w}\widetilde{a}(j)$ will be exclusively produced in (H), while those for which $wa(j) > \widetilde{w}\widetilde{a}(j)$ will be exclusively produced in (F). Equivalently, in

[5] Alternatively, it is sufficient to rule out trade deficits. The absence of a trade surplus would then follow, as consumers' nonsatiability implies that leaving goods unconsumed is incompatible with utility maximization.

terms of $A(j) := \frac{\widetilde{a}(j)}{a(j)}$,

- any $j \in J$ satisfying $\frac{w}{\widetilde{w}} < A(j)$ is produced in (H),
- any $j \in J$ satisfying $\frac{w}{\widetilde{w}} > A(j)$ is produced in (F).

Which country does produce the good j^* that satisfies $\frac{w}{\widetilde{w}} = A(j^*)$? We do not make any assumption in this regard. Good j^* could be produced in (H), (F), or both. Since there is a continuum of goods, the behavior of any specific good has a negligible impact on the economy (technically, the good has zero measure) and thus does not affect equilibrium values. This is in contrast to what happens when there is a discrete number of goods, where the marginal good affects equilibrium.

Once the threshold good j^* is identified, we can define the set of goods produced by each country. Specifically, since $A(j)$ is strictly decreasing, there exists a unique j^* such that

$$\frac{w}{\widetilde{w}} = A(j^*) . \tag{3}$$

Given this,

- country (H) will produce each good $j \in [0, j^*]$,
- country (F) will produce each good $j \in [j^*, 1]$.[6]

While above I have assumed that both countries produce j^*, recall that the identity of the country producing j^* is irrelevant given the continuum of goods.

A key implication of this result is that relative efficiencies fully determine the pattern of production and trade. In other words, countries specialize in producing and exporting goods for which they possess a comparative advantage. Notably, whether a country has an absolute advantage in producing a particular good, as measured by the comparison between $a(j)$ and $\widetilde{a}(j)$, does not influence the allocation of production for that good.

4.2 Demand side

The household's demand of good j in each country is

$$q(j) = \beta(j) \frac{y}{p(j)}$$

[6]While (3) pins down the value of j^* implicitly, we could also obtain an explicit expression. This is based on that A is invertible, because $A(j)$ is decreasing and therefore monotone. Formally, $\exists A^{-1}$ such that $A^{-1} : \mathbb{R}_+ \to Z$, where A^{-1} provides the index of good for a given relative efficiency. Using this, j^* is identified in equilibrium by using $\frac{w}{\widetilde{w}} = A(j^*)$ and hence $j^* = A^{-1}\left(\frac{w}{\widetilde{w}}\right)$.

$$\widetilde{q}(j) = \beta(j)\frac{\widetilde{y}}{p(j)}$$

Given these individual demands, aggregate demand in each country for a good j produced at home is simply

$$Q^{trade}(j) := Lq^{trade}(j) + \widetilde{L}\widetilde{q}^{trade}(j).$$

Note that the expenditure share of each good j will be the same across countries and equal to $\beta(j)$. Formally, $\frac{p(j)q(j)}{y} = \frac{p(j)\widetilde{q}(j)}{\widetilde{y}} = \beta(j)$. Despite this, differences in wages across countries will make the levels of consumption differ across countries. In turn, wages will differ due to differences in technology..

For future references, let

$$B(j^*) := \int_0^{j^*} \beta(j)\ \mathrm{d}j$$

be the fraction of total income spent on goods produced by (H). This implies that

$$1 - B(j^*) = \int_{j^*}^{1} \beta(j)\ \mathrm{d}j$$

is the fraction of total income spent on goods produced by (F). Note that $B'_{j^*} > 0$.

4.3 Balanced Trade

Under open economies, equilibrium requires that both the labor market and the goods market clear in each country. In addition, trade must be balanced, meaning that the value of exports equals the value of imports. This condition ensures that neither country runs a trade surplus or deficit.

To formalize this, recall that the total income of (H) is respectively wL, while that of (F) is $\widetilde{w}\widetilde{L}$. Denote the imports of exports and exports in H as I and X, with a tilde for F. Thus, balanced trade requires $I = X$, or what's the same,

$$\underbrace{[1 - B(j^*)]\,wL}_{(H)\text{'s imports}} = \underbrace{B(j^*)\,\widetilde{w}\widetilde{L}}_{(H)\text{'s exports}}. \tag{4}$$

Rearranging this condition yields the relative wage ratio consistent with balanced trade:

$$\frac{w}{\widetilde{w}} = \frac{B(j^*)}{1 - B(j^*)}\frac{\widetilde{L}}{L}, \tag{5}$$

Remark. There are different ways to arrive at the same equilibrium condition (5).

For instance, we could have used that (H)'s income must equal its production value, where the latter is given by the world demand of (H)'s goods:

$wL = \underbrace{B(j^*)\,wL}_{\text{domestic-demand value}} + \underbrace{B(j^*)\,\widetilde{w}\widetilde{L}}_{\text{foreign-demand value}}$

$\Rightarrow wL = \underbrace{B(j^*)}_{\text{income's share on goods produced at } (H)} \underbrace{\left(wL + \widetilde{w}\widetilde{L}\right)}_{\text{global income}}$

$\Rightarrow \frac{w}{\widetilde{w}} = \frac{B(j^*)}{1-B(j^*)} \frac{\widetilde{L}}{L}$

Alternatively, market-clearing of the labor market provides the same equilibrium condition. For example, considering country H, labor demand must equal labor supply:

$\int_0^{j^*} a(j) \left[Lq(j) + \widetilde{L}\widetilde{q}(j)\right] \mathrm{d}j = L$

$\Rightarrow \int_0^{j^*} a(j) \left[L\beta(j) \frac{w}{p(j)} + \widetilde{L}\widetilde{\beta}(j) \frac{\widetilde{w}}{p(j)}\right] \mathrm{d}j = L.$

Then, substituting in $p(j) = wa(j)$,

$\Rightarrow \int_0^{j^*} \left[L\beta(j) + \widetilde{L}\beta(j) \frac{\widetilde{w}}{w}\right] \mathrm{d}j = L \Rightarrow \left[L + \widetilde{L}\frac{\widetilde{w}}{w}\right] \int_0^{j^*} \beta(j)\, \mathrm{d}j = L$

$\Rightarrow \left[L + \widetilde{L}\frac{\widetilde{w}}{w}\right] B(j^*) = L$

$\Rightarrow \frac{w}{\widetilde{w}} = \frac{B(j^*)}{1-B(j^*)} \frac{\widetilde{L}}{L}$

Note that balanced trade is equivalent to demanding each country's income be equal to its expenditure. Thus, there is no international lending or borrowing. This interpretation follows by adding H's domestic consumption to both sides of (4). Then, the left-hand side would represent H's total expenditure, while the right-hand side would represent its total income. The equivalence will become important for the interpretation of comparative-static results.

4.4 Equilibrium

Overall, all equilibrium values can be obtained by identifying $\frac{w}{\widetilde{w}}$ and j^* through equations (3) and (5). It can be shown that the equilibrium is unique.

For future references, we state these equations again:

$$\frac{w}{\widetilde{w}} = A(j^*),$$

$$\frac{w}{\widetilde{w}} = \frac{B(j^*)}{1 - B(j^*)} \frac{\widetilde{L}}{L}.$$

4.5 Gains of Trade

Next, we prove the existence of gains of trade. This requires comparing welfare per capita under trade, relative to autarky.

Welfare in each scenario is given by

$$\ln v^{aut}(\mathbf{p}, y) := \int_0^1 \beta(j) \ln \left[\frac{\beta(j)}{a(j)}\right] \mathrm{d}j,$$
$$\ln v^{trade}(\mathbf{p}, y) := \int_0^{j^*} \beta(j) \ln \left[\frac{\beta(j)}{a(j)}\right] \mathrm{d}j + \int_{j^*}^1 \beta(j) \ln \left[\frac{\beta(j)}{\widetilde{a}(j)} A(j^*)\right] \mathrm{d}j.$$

implying that gains of trade can be identified through

$$\ln \left[\frac{v^{trade}(\mathbf{p}, y)}{v^{aut}(\mathbf{p}, y)}\right] = \int_{j^*}^1 \beta(j) \ln \left[\frac{A(j^*)}{A(j)}\right] \mathrm{d}j.$$

Since $A(j)$ is decreasing, $\frac{A(j^*)}{A(j)} > 1$ when $j > j^*$. Hence, the left-hand side is positive, implying that $v^{trade}(\mathbf{p}, y) > v^{aut}(\mathbf{p}, y)$ and therefore there are always gains of trade.

Derivation. The expression for $\ln v^{aut}(\mathbf{p}, y)$ was already derived above. Regarding $v^{trade}(\mathbf{p}, y)$, it is given by the utility function (1) evaluated at the vector of consumption $\left(q^{trade}(j)\right)_{j \in [0,1]}$. Since $q^{trade}(j) = \beta(j) \frac{w}{p^{trade}(j)}$, we get

$$\ln v^{trade}(\mathbf{p}, y) = \int_0^1 \beta(j) \ln \left[\beta(j) \frac{w}{p^{trade}(j)}\right] \mathrm{d}j$$

Moreover, we know that $p^{trade}(j) = wa(j)$ if $j \in [0, j^*]$ and $p^{trade}(j) = \widetilde{w}\widetilde{a}(j)$ if $j \in [j^*, 1]$. Therefore,

$$\ln v^{trade}(\mathbf{p}, y) = \int_0^{j^*} \beta(j) \ln \left[\beta(j) \frac{w}{wa(j)}\right] \mathrm{d}j + \int_{j^*}^1 \beta(j) \ln \left[\beta(j) \frac{w}{\widetilde{w}\widetilde{a}(j)}\right] \mathrm{d}j$$

and using that $A(j) := \frac{\widetilde{a}(j)}{a(j)}$ and $\frac{w}{\widetilde{w}} = A(j^*)$, then

$$\ln v^{trade}(\mathbf{p}, y) = \int_0^{j^*} \beta(j) \ln \left[\frac{\beta(j)}{a(j)}\right] \mathrm{d}j + \int_{j^*}^1 \beta(j) \ln \left[\frac{\beta(j)}{\widetilde{a}(j)} A(j^*)\right] \mathrm{d}j$$

Regarding the gains of trade,

$$\begin{aligned}
\ln \left[\frac{v^{trade}(\mathbf{p}, y)}{v^{aut}(\mathbf{p}, y)}\right] &= \int_0^{j^*} \beta(j) \ln \left[\frac{\beta(j)}{a(j)}\right] \mathrm{d}j + \int_{j^*}^1 \beta(j) \ln \left[\frac{\beta(j)}{\widetilde{a}(j)} A(j^*)\right] \mathrm{d}j - \int_0^1 \beta(j) \ln \left(\frac{\beta(j)}{a(j)}\right) \mathrm{d}j, \\
&= \int_{j^*}^1 \beta(j) \ln \left[\frac{\beta(j)}{\widetilde{a}(j)} A(j^*)\right] \mathrm{d}j - \int_{j^*}^1 \beta(j) \ln \left(\frac{\beta(j)}{a(j)}\right) \mathrm{d}j, \\
&= \int_{j^*}^1 \beta(j) \ln \left[\frac{\beta(j)}{\widetilde{a}(j)} A(j^*) \frac{a(j)}{\beta(j)}\right] \mathrm{d}j, \\
&= \int_{j^*}^1 \beta(j) \ln \left[\frac{A(j^*)}{A(j)}\right] \mathrm{d}j.
\end{aligned}$$

4.6 The Role of Absolute Advantages

Previously, we showed that the pattern of production and trade is determined by relative efficiencies. They are embodied in $A(j)$ and form the basis of comparative advantage. In contrast, the level of efficiencies $a(j)$ and $\widetilde{a}(j)$ do not influence these patterns when considered in isolation. Thus, a country could possess absolute advantages in every good, without this indicating which goods it will produce and export.

While absolute advantages do not dictate trade patterns, they may still be relevant for other outcomes within the model. In particular, **a country with absolute advantages**

in all goods will necessarily have a greater income. Formally, country (H) has absolute advantages in all goods when $a(j) < \widetilde{a}(j)$ for every $j \in J$, so that $A(j) > 1$. The result follows directly from (3) and using $A(j^*) > 1$.

In a context where the law of one price holds, the result immediately implies that country (H) would also enjoy a greater welfare relative to country (F). Depending on the analysis, the result could be of significance. For instance, it determines that (F) would have incentives to enhance its technology in a way that it becomes the country with absolute advantages in every good.

5 Free Trade: Differences in Market Size

Having established that trade benefits all countries, we can further explore the implications of the model by conducting a **comparative-static analysis**. This requires shocking one parameter and then examining how the endogenous variables respond to this change.

5.1 Results

In particular, let's consider an increase in the population of (F). Formally, we consider an infinitesimal variation $\mathrm{d}\widetilde{L} > 0$. Our interest lies in the signs of $\frac{\partial(w/\widetilde{w})}{\partial\widetilde{L}}$ and $\frac{\partial j^*}{\partial\widetilde{L}}$.

Differentiating the system of equilibrium conditions given by (3) and (5), and expressing it in a matrix form:

$$\begin{pmatrix} 1 & -A'(j^*) \\ 1 & \frac{-B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L} \end{pmatrix} \begin{pmatrix} \mathrm{d}(w/\widetilde{w}) \\ \mathrm{d}j^* \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{B(j^*)}{1-B(j^*)} \end{pmatrix} \mathrm{d}\widetilde{L}.$$

Dividing both sides by $\mathrm{d}\widetilde{L}$, we obtain our expression of interest:

$$\begin{pmatrix} 1 & -A'(j^*) \\ 1 & \frac{-B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L} \end{pmatrix} \begin{pmatrix} \frac{\partial(w/\widetilde{w})}{\partial\widetilde{L}} \\ \frac{\partial j^*}{\partial\widetilde{L}} \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{B(j^*)}{1-B(j^*)} \end{pmatrix}.$$

Solving the system, we can establish that an increase in the population of (F) relative to (H) results in:

- $\frac{\partial(w/\widetilde{w})}{\partial\widetilde{L}} > 0$: the relative wage of (H) increases.
- $\frac{\partial j^*}{\partial\widetilde{L}} < 0$: (H) produces and exports fewer goods, while (F) produces and exports

more goods.

Derivation. The equilibrium conditions are $\frac{w}{\widetilde{w}} = A(j^*)$ and $\frac{w}{\widetilde{w}} = \frac{B(j^*)}{1-B(j^*)}\frac{\widetilde{L}}{L}$. We differentiate each of them in terms of the endogenous variables and the exogenous variable of interest, $\mathrm{d}\widetilde{L} \neq 0$. Formally, our goal is to study the impact of $\mathrm{d}\widetilde{L} \neq 0$ on $\mathrm{d}(w/\widetilde{w})$ and $\mathrm{d}j^*$. Differentiating each equation:

$$\mathrm{d}(w/\widetilde{w}) = A'(j^*)\,\mathrm{d}j^* + 0\,\mathrm{d}\widetilde{L},$$

$$\mathrm{d}(w/\widetilde{w}) = \left[\frac{B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L}\right]\mathrm{d}j^* + \left[\frac{B(j^*)}{1-B(j^*)}\right]\mathrm{d}\widetilde{L}.$$

In matrix form, we can express these equations as in the main text:

$$\begin{pmatrix} 1 & -A'(j^*) \\ 1 & \frac{-B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L} \end{pmatrix}\begin{pmatrix} \frac{\partial(w/\widetilde{w})}{\partial\widetilde{L}} \\ \frac{\partial j^*}{\partial\widetilde{L}} \end{pmatrix} = \begin{pmatrix} 0 \\ \frac{B(j^*)}{1-B(j^*)} \end{pmatrix}$$

The matrix on the left-hand side corresponds to the Jacobian matrix. Formally, let $J := \begin{pmatrix} 1 & -A'(j^*) \\ 1 & \frac{-B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L} \end{pmatrix}$ with $\det J = \frac{[1-B(j^*)]^2 A'(j^*) - B'(j^*)\frac{\widetilde{L}}{L}}{[1-B(j^*)]^2}$. Note that $\det J < 0$.

Solving the system by Cramer's rule:

$$\begin{aligned}\frac{\partial(w/\widetilde{w})}{\partial\widetilde{L}} &= \frac{\det\begin{pmatrix} 0 & -A'(j^*) \\ \frac{B(j^*)}{1-B(j^*)} & \frac{B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L} \end{pmatrix}}{\det J}, \\ &= \frac{A'(j^*)}{\det J}\frac{B(j^*)}{1-B(j^*)}, \\ &> 0.\end{aligned}$$

and

$$\begin{aligned}\frac{\partial j^*}{\partial\widetilde{L}} &= \frac{\det\begin{pmatrix} 1 & 0 \\ 1 & \frac{B(j^*)}{1-B(j^*)}\frac{\widetilde{L}}{L} \end{pmatrix}}{\det J}, \\ &= \frac{1}{\det J}\frac{B(j^*)}{1-B(j^*)}, \\ &< 0.\end{aligned}$$

5.2 Interpretation

There are several ways to interpret the results obtained. To begin, let us restate the findings from the perspective of country F: wages in F fall relative to wages in H, and F increases the number of goods exported.

One interpretation emphasizes labor market dynamics. As noted earlier, the balanced-trade condition is equivalent to the equality of labor supply and demand. When the population in F rises, the supply of labor exceeds demand, which reduces wages in F relative to H. Lower wages make goods produced in F cheaper, thereby inducing an increase in exports from F.

A second interpretation introduces the role of exchange rates in wage adjustment. Assume a flexible exchange rate, where e denotes the price of foreign currency in terms of H's currency. Under this definition, an increase in e represents a depreciation of H's currency, while a decrease in e corresponds to an appreciation of H's currency. Wages at home relative to F, expressed in units of home currency, can be written as

$$\frac{w}{\widetilde{w}} = \frac{W}{e\widetilde{W}}.$$

where W and $\widetilde{W}$ are wages expressed in the currencies of the own country.

If wages are sticky, both W and $\widetilde{W}$ remain fixed. Thus, an increase in relative home wages occurs when e decreases. This implies an appreciation in H and, equivalently, a depreciation in F. Overall, an increase in F's population leads to a depreciation of F's currency and a rise in its exports.

This mechanism can be understood as follows. A larger population in F initially raises imports, which increases the demand for H's currency in F. This higher demand for H's currency causes F's currency to depreciate. The depreciation makes goods from F cheaper when priced in H's currency, encouraging H to import more from F. Consequently, F's exports expand.

5.3 Impact on Welfare

Once we have identified the impact of an increase in $\widetilde{L}$, we can also evaluate how this shock affects welfare per capita at home. This is given by

$$\frac{\mathrm{d}\ln v^{trade}(\mathbf{p},y)}{\mathrm{d}\widetilde{L}} = \frac{\partial \ln v^{trade}(\mathbf{p},y)}{\partial j^*}\frac{\partial j^*}{\partial \widetilde{L}} > 0,$$

implying that increases in $\widetilde{L}$ (either a greater population of (F) and/or a lower population of (H) benefits (H).

Derivation. Welfare is given by

$$\ln v^{trade}(\mathbf{p},y) = \int_0^{j^*} \beta(j)\ln\left[\frac{\beta(j)}{a(j)}\right]\mathrm{d}j + \int_{j^*}^1 \beta(j)\ln\left[\frac{\beta(j)}{\widetilde{a}(j)}A(j^*)\right]\mathrm{d}j.$$

The expression does not depend directly on $\widetilde{L}$, but rather through j^*. Formally, this implies that:

$$\frac{\mathrm{d}\ln v^{trade}(\mathbf{p},y)}{\mathrm{d}\widetilde{L}} = \frac{\partial \ln v^{trade}(\mathbf{p},y)}{\partial j^*}\frac{\partial j^*}{\partial \widetilde{L}}$$

Through the comparative static analysis, we have already determined that $\frac{\partial j^*}{\partial \widetilde{L}} = \frac{B(j^*)[1-B(j^*)]}{[1-B(j^*)]^2 A'(j^*) - B'(j^*)} < 0$. The sign of $\frac{\partial \ln v^{trade}(\mathbf{p},y)}{\partial j^*}$ remains to be determined:

$$\begin{aligned}
\frac{\partial \ln v^{trade}(\mathbf{p},y)}{\partial j^*} &= \beta(j^*)\ln\left[\frac{\beta(j^*)}{a(j^*)}\right] - \beta(j^*)\ln\left[\frac{\beta(j^*)}{\widetilde{a}(j^*)}A(j^*)\right] + \int_{j^*}^{1}\beta(j)\,\frac{A'(j^*)}{A(j^*)}\mathrm{d}j,\\
&= \beta(j^*)\ln\left[\frac{\beta(j^*)}{a(j^*)}\right] - \beta(j^*)\ln\left[\frac{\beta(j^*)}{a(j^*)}\right] + \frac{A'(j^*)}{A(j^*)}\int_{j^*}^{1}\beta(j)\,\mathrm{d}j,\\
&= \frac{A'(j^*)}{A(j^*)}\left[1-B(j^*)\right] < 0.
\end{aligned}$$

Therefore,

$$\frac{\mathrm{d}\ln v^{trade}(\mathbf{p},y)}{\mathrm{d}\widetilde{L}} = \underbrace{\frac{\partial \ln v^{trade}(\mathbf{p},y)}{\partial j^*}}_{-}\underbrace{\frac{\partial j^*}{\partial \widetilde{L}}}_{-} > 0$$

The term $\frac{\mathrm{d}\ln v^{trade}(\mathbf{p},y)}{\mathrm{d}\widetilde{L}}$ captures the increase in welfare when (H) is already trading with (F). But this coincides with the change in *gains of trade*, given by $\ln\left(\frac{v^{trade}(\mathbf{p},y)}{v^{aut}(\mathbf{p},y)}\right)$. To see this, note that

$$\begin{aligned}
\frac{\mathrm{d}\ln\left(\frac{v^{trade}(\mathbf{p},y)}{v^{aut}(\mathbf{p},y)}\right)}{\mathrm{d}\widetilde{L}} &= \frac{\partial\ln\left(\frac{v^{trade}(\mathbf{p},y)}{v^{aut}(\mathbf{p},y)}\right)}{\partial j^*}\frac{\partial j^*}{\partial \widetilde{L}},\\
&= \left[\frac{\partial \ln\left(v^{trade}(\mathbf{p},y)\right)}{\partial j^*} - \underbrace{\frac{\partial \ln\left(v^{aut}(\mathbf{p},y)\right)}{\partial j^*}}_{=0}\right]\frac{\partial j^*}{\partial \widetilde{L}},\\
&= \frac{\partial \ln\left(v^{trade}(\mathbf{p},y)\right)}{\partial j^*}\frac{\partial j^*}{\partial \widetilde{L}},\\
&= \frac{\mathrm{d}\ln\left(v^{trade}(\mathbf{p},y)\right)}{\mathrm{d}\widetilde{L}}.
\end{aligned}$$

6 Free Trade: Improvements in Technology of the Trading Partner

In the following, we consider how improvements in technology in the foreign country affect welfare in H. In particular, we show that productivity gains in a trading partner can either raise or lower welfare in H. Moreover, we provide intuition for the conditions under which each outcome arises.

6.1 Result

We start by considering general improvements in F's technology. After this, we will restrict attention to specific cases.

Let the unit labor requirements in F be parametrized as $\widetilde{a}(j;\delta)$, where $\frac{\partial \widetilde{a}(j)}{\partial \delta} < 0$. Thus, increases in δ reduce the labor required to produce good j. The relative efficiency is now

$$A(j) := \frac{\widetilde{a}(j;\delta)}{a(j)},$$

where we assume variations in δ affect relative efficiencies, but preserve the ranking of goods. Formally, $A'_j(j;\delta) < 0$ for any δ.

Performing comparative statics with respect to δ, we obtain:

$$\frac{\partial (w/\widetilde{w})}{\partial \delta} = \frac{-\Theta A^{*\prime}_{\delta}}{A^{*\prime}_{j} - \Theta} < 0$$

$$\frac{\partial j^*}{\partial \delta} = \frac{-A^{*\prime}_{\delta}}{A^{*\prime}_{j} - \Theta} < 0$$

where $A^{*\prime}_{\delta} := \frac{\partial A(j^*;\delta)}{\partial \delta}$ and $A^{*\prime}_{j} := \frac{\partial A(j^*;\delta)}{\partial j}$.

Derivation. Differentiating the equilibrium conditions:

$$\mathrm{d}(w/\widetilde{w}) = \frac{\partial A(j^*;\delta)}{\partial j^*}\mathrm{d}j^* + \frac{\partial A(j^*;\delta)}{\partial \delta}\mathrm{d}\delta$$

$$\mathrm{d}(w/\widetilde{w}) = \left(\frac{B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L}\right)\mathrm{d}j^*$$

In matrix form, we can express these equations as in the main text:

$$\begin{pmatrix} 1 & -\frac{\partial A(j^*;\delta)}{\partial j^*} \\ 1 & \frac{-B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L} \end{pmatrix} \begin{pmatrix} \mathrm{d}(w/\widetilde{w}) \\ \mathrm{d}j^* \end{pmatrix} = \begin{pmatrix} \frac{\partial A(j^*;\delta)}{\partial \delta} \\ 0 \end{pmatrix}\mathrm{d}\delta$$

The matrix on the left-hand side corresponds to the Jacobian matrix. Formally, let $J := \begin{pmatrix} 1 & -A'_j(j^*;\delta) \\ 1 & -\Theta \end{pmatrix}$ where $\Theta := \frac{B'(j^*)}{[1-B(j^*)]^2}\frac{\widetilde{L}}{L}$. Then, with $\det J = A'_{j^*} - \Theta$ where $\det J < 0$.

Solving the system by Cramer's rule:

$$\frac{\partial (w/\widetilde{w})}{\partial \delta} = \frac{\det\begin{pmatrix} \frac{\partial A(j^*;\delta)}{\partial \delta} & -\frac{\partial A(j^*;\delta)}{\partial j^*} \\ 0 & -\Theta \end{pmatrix}}{\det J} = \frac{-\Theta\frac{\partial A(j^*;\delta)}{\partial \delta}}{\det J} < 0$$

$$\frac{\partial j^*}{\partial \delta} = \frac{\det\begin{pmatrix} 1 & \frac{\partial A(j^*;\delta)}{\partial \delta} \\ 1 & 0 \end{pmatrix}}{\det J} = \frac{-\frac{\partial A(j^*;\delta)}{\partial \delta}}{\det J} < 0$$

We have already obtained welfare per capita in H:

$$\ln v^{trade}(\mathbf{p}, y) = \int_0^{j^*} \beta(j)\ln\left[\frac{\beta(j)}{a(j)}\right]\mathrm{d}j + \int_{j^*}^1 \beta(j)\ln\left[\frac{\beta(j)A(j^*;\delta)}{\widetilde{a}(j;\delta)}\right]\mathrm{d}j$$

Following an increase in δ, the impact on H's welfare is

$$\frac{\mathrm{d}\ln v^{trade}}{\mathrm{d}\delta} = \frac{\partial \ln v^{trade}}{\partial \delta} + \frac{\partial \ln v^{trade}}{\partial j^*}\frac{\partial j^*}{\partial \delta},$$

which equals

$$\frac{\mathrm{d}\ln v^{trade}}{\mathrm{d}\delta}=\left[1-B\left(j^{*}\right)\right]\left[\frac{-\Theta}{A'_{j^{*}}-\Theta}\right]\frac{A'_{\delta}\left(j^{*};\delta\right)}{A\left(j^{*}\right)}-\int_{j^{*}}^{1}\beta\left(j\right)\frac{A'_{\delta}\left(j;\delta\right)}{A\left(j\right)}\mathrm{d}j \tag{6}$$

Derivation. Use that

$$A'_{j}\left(j^{*};\delta\right)=\frac{\widetilde{a}'_{j^{*}}a\left(j^{*}\right)-\widetilde{a}\left(j^{*}\right)a'_{j^{*}}}{\left[a\left(j^{*}\right)\right]^{2}}$$
$$\Rightarrow\frac{A'_{j}\left(j^{*};\delta\right)}{A\left(j^{*}\right)}=\frac{\partial\ln\widetilde{a}\left(j^{*};\delta\right)}{\partial j}-\frac{\partial\ln a\left(j^{*};\delta\right)}{\partial j}$$

and

$$A'_{\delta}\left(j;\delta\right)=\frac{\widetilde{a}'_{\delta}\left(j\right)}{a\left(j\right)}$$
$$\Rightarrow\frac{A'_{\delta}\left(j;\delta\right)}{A\left(j\right)}=\frac{\partial\ln\widetilde{a}\left(j;\delta\right)}{\partial\delta}$$

Previously, we have shown that

$$\frac{\partial\ln v^{trade}}{\partial j^{*}}=\frac{A'_{j}\left(j^{*};\delta\right)}{A\left(j^{*}\right)}\left[1-B\left(j^{*}\right)\right],$$

Moreover,

$$\begin{aligned}\frac{\partial\ln v^{trade}}{\partial\delta}&=\left[1-B\left(j^{*}\right)\right]\frac{\partial\ln\widetilde{a}\left(j^{*};\delta\right)}{\partial\delta}-\int_{j^{*}}^{1}\beta\left(j\right)\frac{\partial\ln\widetilde{a}\left(j;\delta\right)}{\partial\delta}\mathrm{d}j,\\&=\left[1-B\left(j^{*}\right)\right]\frac{A'_{\delta}\left(j^{*};\delta\right)}{A\left(j^{*}\right)}-\int_{j^{*}}^{1}\beta\left(j\right)\frac{A'_{\delta}\left(j;\delta\right)}{A\left(j\right)}\mathrm{d}j\end{aligned}$$

Substituting, we get the result.

We now restrict the analysis to specific technological improvements in F. They differ on which goods are affected.

6.2 Increases in H's Welfare

Suppose that technological improvements in country F occur in goods where F already holds a comparative advantage. Formally, while $A'_{\delta}\left(j;\delta\right)<0$ for $j\in\left[j^{*},1\right]$, there is a negligible effect for the rest of goods: $A'_{\delta}\left(j;\delta\right)\to0$ for $j\in\left[0,j^{*}\right]$. Note that the marginal good remains unaffected under these assumptions.

Then, the impact on welfare (6) is

$$\begin{aligned}\frac{\mathrm{d}\ln v^{trade}}{\mathrm{d}\delta}&=\underbrace{\left[1-B\left(j^{*}\right)\right]\left[\frac{-\Theta}{A'_{j^{*}}-\Theta}\right]\frac{A'_{\delta}\left(j^{*};\delta\right)}{A\left(j^{*}\right)}}_{=0}-\int_{j^{*}}^{1}\beta\left(j\right)\frac{A'_{\delta}\left(j;\delta\right)}{A\left(j\right)}\mathrm{d}j,\\&=-\int_{j^{*}}^{1}\beta\left(j\right)\frac{A'_{\delta}\left(j;\delta\right)}{A\left(j\right)}\mathrm{d}j,\\&>0.\end{aligned}$$

Thus, H experiences a welfare gain.

To interpret the result, consider equilibrium conditions (3) and (5). They establish that only changes in productivity that affect the threshold good can influence either $\frac{w}{\widetilde{w}}$ or j^*. Consequently, improvements in F's technology do not alter relative wages in H. Instead, they reduce the cost of H's imports, which raises real income per capita.

6.3 Decreases in H's Welfare

Suppose now that the technological improvement in F affect goods in which H has comparative advantages. Formally, $A'_\delta(j;\delta) \to 0$ for $j \in [j^*, 1]$, so that productivity changes are negligible for goods $j \in [j^*, 1]$. Instead, $A'_\delta(j;\delta) < 0$ for $j \in [0, j^*]$. Now, the marginal good is affected.

The impact on welfare determined by (6) becomes

$$\begin{aligned}\frac{\mathrm{d}\ln v^{trade}}{\mathrm{d}\delta} &= \left[1 - B(j^*)\right]\left[\frac{-\Theta}{A'_{j^*} - \Theta}\right]\frac{A'_\delta(j^*;\delta)}{A(j^*)} - \underbrace{\int_{j^*}^{1} \beta(j)\frac{A'_\delta(j;\delta)}{A(j)}\mathrm{d}j}_{=0}, \\ &= \left[1 - B(j^*)\right]\left[\frac{-\Theta}{A'_{j^*} - \Theta}\right]\frac{A'_\delta(j^*;\delta)}{A(j^*)}, \\ &< 0.\end{aligned}$$

Thus, H is worse off.

The key is that F's improvement in technology reduce relative wages in H. In particular, we have previously shown that $\frac{\partial(w/\widetilde{w})}{\partial\delta} < 0$ and $\frac{\partial j^*}{\partial\delta} < 0$. To build intuition, note that H will continue to produce the bulk of the goods in which it has a comparative advantage. Only those in a neighborhood around j^* are affected. Thus, the price paid by consumers in H is only affected for a reduced set of goods: those near j^* that shift to being imported from F. Instead, the impact on relative wages due to changes in $A(j^*)$ have a broad impact on welfare, since it reduces the purchasing power of consumers across all goods. In other words, the fall in relative wages implies that H's households can afford less overall consumption, leading to a decline in real income.

We can consider a more general technological improvement in F. A necessary condition for trading partner's technological improvements hurt home is that $\frac{\partial^2 \ln A(s;\delta)}{\partial\delta\partial s} > 0$. Expressed in words, the technological improvements affect all goods, but are more pronounced for goods in which H has comparative advantages.

To see this why this is necessary, note that

$$\begin{aligned}\frac{\mathrm{d}\ln v^{trade}}{\mathrm{d}\delta} &= \left[1-B\left(j^*\right)\right]\left[\frac{-\Theta}{A'_{j^*}-\Theta}\right]\frac{A'_\delta\left(j^*;\delta\right)}{A\left(j^*\right)}-\int_{j^*}^1\beta\left(j\right)\frac{A'_\delta\left(j;\delta\right)}{A\left(j\right)}\mathrm{d}j,\\ &= \left[\frac{-\Theta}{A'_{j^*}-\Theta}\right]\int_{j^*}^1\beta\left(j\right)\frac{A'_\delta\left(j^*;\delta\right)}{A\left(j^*\right)}\mathrm{d}j-\int_{j^*}^1\beta\left(j\right)\frac{A'_\delta\left(j;\delta\right)}{A\left(j\right)}\mathrm{d}j,\\ &= \int_{j^*}^1\beta\left(j\right)\left[\frac{A'_\delta\left(j^*;\delta\right)}{A\left(j^*\right)}\kappa-\frac{A'_\delta\left(j;\delta\right)}{A\left(j\right)}\right]\mathrm{d}j.\end{aligned}$$

where $\kappa^* := \frac{-\Theta}{A'_{j^*}-\Theta} \in (0,1)$ is a constant. By the fundamental theorem of calculus: $\frac{\partial \ln A(j^*;\delta)}{\partial\delta}-\frac{\partial \ln A(j;\delta)}{\partial\delta} = -\int_{j^*}^{j}\frac{\partial^2\ln A(s;\delta)}{\partial\delta\partial s}\mathrm{d}s.$

$$\begin{aligned}\frac{\mathrm{d}\ln v^{trade}}{\mathrm{d}\delta} &= \int_{j^*}^1\beta\left(j\right)\left[\frac{\partial\ln A\left(j^*;\delta\right)}{\partial\delta}\kappa^*-\frac{\partial\ln A\left(j;\delta\right)}{\partial\delta}\right]\mathrm{d}j,\\ &= \int_{j^*}^1\beta\left(j\right)\left[\frac{\partial\ln A\left(j^*;\delta\right)}{\partial\delta}\kappa^*+\frac{\partial\ln A\left(j^*;\delta\right)}{\partial\delta}-\frac{\partial\ln A\left(j^*;\delta\right)}{\partial\delta}-\frac{\partial\ln A\left(j;\delta\right)}{\partial\delta}\right]\mathrm{d}j,\\ &= \int_{j^*}^1\beta\left(j\right)\left[\frac{\partial\ln A\left(j^*;\delta\right)}{\partial\delta}\left(\kappa^*-1\right)-\int_{j^*}^{j}\frac{\partial^2\ln A\left(s;\delta\right)}{\partial\delta\partial s}\mathrm{d}s\right]\mathrm{d}j.\end{aligned}$$

Since $\frac{\partial\ln A(j^*;\delta)}{\partial\delta}<0$ and $\kappa^*<1$, we need that $\frac{\partial^2\ln A(j;\delta)}{\partial\delta\partial j}>0$ to possibly obtain $\frac{\mathrm{d}\ln v^{trade}}{\mathrm{d}\delta}<0$.

7 Costly Trade and Nontradable goods

Economies rarely operate at the extremes of autarky or free trade. A more realistic representation accounts for frictions arising when goods are exchanged across borders, which can be more or less significant. Next, we consider this scenario by introducing the concept of **trade costs**: additional expenses relative to domestic delivery, to make a good available in a foreign market.

From a technical perspective, the incorporation of trade costs leads to a more general model, where autarky and free trade emerge as special cases. Specifically, autarky corresponds to a scenario where trade costs are prohibitively high, effectively making it impossible to trade with other countries. On the other hand, free trade emerges as a scenario where trade costs are zero.

There are different ways to model trade costs. We adopt the standard form of the so-called **iceberg transportation costs**, which considers that a physical portion of the good is lost during transportation. The term "iceberg" is reminiscent of an iceberg melting away when it is transported.

Formally, iceberg costs determine that only $\frac{1}{\tau}$ with $\tau>1$ arrives when one unit is sent. Consequently, the firm needs to ship τ units to ensure that one unit is consumed abroad. The existence of trade costs creates a **wedge between what the firm produces**

and what the consumer gets. Furthermore, **the law of one price does not hold anymore**. In particular, a consumer pays different prices depending on whether the good is produced by a domestic or a foreign firm. This also determines that some goods are produced solely for domestic consumption, resulting in a range of non-traded goods.

Formally, the price of good j in (H) is still given by the minimum between the domestic or imported alternative. However, foreign firms now must send τ units of good j if the good is imported. This makes the marginal cost of the imported good become $\widetilde{w}\widetilde{a}\,(j)\,\tau$. Instead, the marginal cost to sell domestically is unaffected, remaining the same as in the baseline model. Specifically, the price of good j in (H) is

$$p\,(j) = \inf\left\{wa\,(j)\,, \widetilde{w}\widetilde{a}\,(j)\,\tau\right\}.$$

Consumers in (H) will opt for local production when $wa\,(j) < \widetilde{w}\widetilde{a}\,(j)\,\tau$ or, equivalently,

$$\frac{w}{\widetilde{w}} < A\,(j)\,\tau.$$

This allows us to define a cutoff good j^* in country (H) such that a consumer from (H) is indifferent between buying the good locally or importing it. Formally, j^* satisfies that

$$\frac{w}{\widetilde{w}} = A\,(j^*)\,\tau. \tag{7}$$

By the same token, the price of good j in (F) is

$$\widetilde{p}\,(j) = \inf\left\{wa\,(j)\,\tau, \widetilde{w}\widetilde{a}\,(j)\right\}.$$

So, consumers in (F) choose importing from (H) when

$$\frac{w}{\widetilde{w}} < \frac{A\,(j)}{\tau}.$$

This relation defines the cutoff good $\widetilde{j}^*$ in country (F) such that a consumer from (F) is indifferent between buying the good locally or importing it. Formally,

$$\frac{w}{\widetilde{w}} = \frac{A\left(\widetilde{j}^*\right)}{\tau}. \tag{8}$$

Notice that (7) and (8) make it possible to define when (H) is producing exclusively for its home market or when it additionally exports the good. This occurs since (H) exporting a good means that (F) is importing it. Therefore, (H) produces goods for both the local and export market when both inequalities hold:

$$\frac{w}{\widetilde{w}} < A(j)\,\tau,$$
$$\frac{w}{\widetilde{w}} < \frac{A(j)}{\tau}.$$

Since $\tau > 1$ and both inequalities have to hold, then good j will be consumed locally and exported if $\frac{w}{\widetilde{w}} < \frac{A(j)}{\tau}$.

In addition, (F) will produce for both countries when

$$\frac{w}{\widetilde{w}} > A(j)\,\tau,$$
$$\frac{w}{\widetilde{w}} > \frac{A(j)}{\tau},$$

which implies that (F) will produce for the home country and export good j when $\frac{w}{\widetilde{w}} > A(j)\,\tau$.

The relation between j^* and $\widetilde{j}^*$ becomes important to describe the pattern of trade. In particular, it can be shown that $j^* > \widetilde{j}^*$.

Derivation. By definition, j^* is such that $\frac{w}{\widetilde{w}} = A(j^*)\,\tau$ and $\widetilde{j}^*$ such that $\frac{w}{\widetilde{w}} = \frac{A(\widetilde{j}^*)}{\tau}$. These two equations imply that:

$$A(j^*)\,\tau = \frac{A\left(\widetilde{j}^*\right)}{\tau}.$$

Consequently, $A(j^*)\,\tau^2 = A\left(\widetilde{j}^*\right)$. From this we deduce that $A(j^*) < A\left(\widetilde{j}^*\right)$, since $\tau > 1$. In addition, A was defined such that goods were ordered in a way that $A' < 0$. Thus, $A(j^*) < A\left(\widetilde{j}^*\right)$ can only happen if $j^* > \widetilde{j}^*$.

Since $j^* > \widetilde{j}^*$, we can determine that

$$j \in \begin{cases} \left[0, \widetilde{j}^*\right] & (H) \text{ serves both } (H) \text{ and } (F) \\ \left[\widetilde{j}^*, j^*\right] & \text{non-traded goods (only sold domestically)} \\ [j^*, 1] & (F) \text{ serves both } (F) \text{ and } (H) \end{cases}$$

which completely describes the patterns of trade.

Notice we only have partially described the equilibrium. This is reflected in that the cutoff goods are still a function of the equilibrium relative wages. These cutoffs are endogenous, and we can use the balanced-trade condition to identify them:

$$\underbrace{[1 - B(j^*)]\,wL}_{\text{home qountry's imports}} = \underbrace{v\left(\widetilde{j}^*\right)\widetilde{w}\widetilde{L}}_{\text{home country's exports}},$$

which can be reexpressed as $\frac{w}{\widetilde{w}} = \frac{B\left(\widetilde{j}^{*}\right)}{1-B(j^{*})}\frac{\widetilde{L}}{L}$.

In sum, all these derivations imply that the system of equilibrium conditions is given by

$$\frac{w}{\widetilde{w}} = \frac{A\left(\widetilde{j}^{*}\right)}{\tau}, \tag{9a}$$
$$\frac{w}{\widetilde{w}} = A\left(j^{*}\right)\tau, \tag{9b}$$
$$\frac{w}{\widetilde{w}} = \frac{B\left(\widetilde{j}^{*}\right)}{1-B\left(j^{*}\right)}\frac{\widetilde{L}}{L}, \tag{9c}$$

As a corollary, the system (9) allows us to pin down $\frac{w}{\widetilde{w}}$, j^{*}, and $\widetilde{j}^{*}$.

Given this, we can revisit what occurs when there are variations in the relative sizes of countries. By differentiating the system and expressing it in a matrix form:

$$\begin{pmatrix} 1 & -\frac{B'_{j^{*}}B\left(\widetilde{j}^{*}\right)}{[1-B(j^{*})]^{2}}\frac{\widetilde{L}}{L} & \frac{-B'_{\widetilde{j}^{*}}}{1-B(j^{*})}\frac{\widetilde{L}}{L} \\ 1 & -A'_{j^{*}}\tau & 0 \\ 1 & 0 & \frac{-A'_{\widetilde{j}^{*}}}{\tau} \end{pmatrix}\begin{pmatrix} \mathrm{d}\left(w/\widetilde{w}\right) \\ \mathrm{d}j^{*} \\ \mathrm{d}\widetilde{j}^{*} \end{pmatrix} = \begin{pmatrix} \frac{B\left(\widetilde{j}^{*}\right)}{1-B(j^{*})} \\ 0 \\ 0 \end{pmatrix}\mathrm{d}\widetilde{L}.$$

By performing comparative statics, we get that

$$\frac{\partial\left(w/\widetilde{w}\right)}{\partial\widetilde{L}} > 0,$$
$$\frac{\partial j^{*}}{\partial\widetilde{L}} < 0,$$
$$\frac{\partial\widetilde{j}^{*}}{\partial\widetilde{L}} < 0.$$

Derivation. We denote the Jacobian of the left-hand side's matrix by Jb. Formally,

$$Jb := \begin{pmatrix} 1 & -\frac{B'_{j^{*}}B\left(\widetilde{j}^{*}\right)}{[1-B(j^{*})]^{2}}\frac{\widetilde{L}}{L} & -\frac{B'_{\widetilde{j}^{*}}}{1-B(j^{*})}\frac{\widetilde{L}}{L} \\ 1 & -A'_{j^{*}}\tau & 0 \\ 1 & 0 & \frac{-A'_{\widetilde{j}^{*}}}{\tau} \end{pmatrix}.$$

First we show that $\det Jb > 0$:

$$\det\begin{pmatrix} 1 & -\frac{B'_{j^{*}}B\left(\widetilde{j}^{*}\right)}{[1-B(j^{*})]^{2}}\frac{\widetilde{L}}{L} & -\frac{B'_{\widetilde{j}^{*}}}{1-B(j^{*})}\frac{\widetilde{L}}{L} \\ 1 & -A'_{j^{*}}\tau & 0 \\ 1 & 0 & \frac{-A'_{\widetilde{j}^{*}}}{\tau} \end{pmatrix} = \underbrace{A'_{j^{*}}A'_{\widetilde{j}^{*}}}_{+} - \underbrace{\frac{B'_{j^{*}}B\left(\widetilde{j}^{*}\right)}{\left[1-B\left(j^{*}\right)\right]^{2}}\frac{\widetilde{L}}{L}\frac{A'_{\widetilde{j}^{*}}}{\tau}}_{-} - \underbrace{\frac{B'_{\widetilde{j}^{*}}}{1-B\left(j^{*}\right)}\frac{\widetilde{L}}{L}A'_{j^{*}}\tau}_{-} > 0$$

where we have used that $A'_{j^{*}}, A'_{\widetilde{j}^{*}} < 0$ and $B'_{j^{*}}, B'_{\widetilde{j}^{*}} > 0$.

Hence,

$$
\begin{aligned}
\frac{\partial\left(w/\widetilde{w}\right)}{\partial\widetilde{L}} &= \left(\det Jb\right)^{-1}\det\begin{pmatrix}
\frac{B\left(\widetilde{j}^{*}\right)}{1-B\left(j^{*}\right)} & -\frac{B'_{j^{*}}B\left(\widetilde{j}^{*}\right)}{\left[1-B\left(j^{*}\right)\right]^{2}}\frac{\widetilde{L}}{L} & -\frac{B'_{\widetilde{j}^{*}}}{1-B\left(j^{*}\right)}\frac{\widetilde{L}}{L}\\
0 & -A'_{j^{*}}\tau & 0\\
0 & 0 & \frac{-A'_{\widetilde{j}^{*}}}{\tau}
\end{pmatrix},\\
&= \left(\det Jb\right)^{-1}\frac{B\left(\widetilde{j}^{*}\right)}{1-B\left(j^{*}\right)}A'_{j^{*}}A'_{\widetilde{j}^{*}},\\
&> 0
\end{aligned}
$$

. Additionally,

$$
\begin{aligned}
\frac{\partial j^{*}}{\partial\widetilde{L}} &= \left(\det Jb\right)^{-1}\det\begin{pmatrix}
1 & \frac{B\left(\widetilde{j}^{*}\right)}{1-B\left(j^{*}\right)} & -\frac{B'_{\widetilde{j}^{*}}}{1-B\left(j^{*}\right)}\frac{\widetilde{L}}{L}\\
1 & 0 & 0\\
1 & 0 & \frac{-A'_{\widetilde{j}^{*}}}{\tau}
\end{pmatrix},\\
&= \left(\det Jb\right)^{-1}\left(-1\right)\frac{B\left(\widetilde{j}^{*}\right)}{1-B\left(j^{*}\right)}\frac{-A'_{\widetilde{j}^{*}}}{\tau},\\
&< 0.
\end{aligned}
$$

And

$$
\begin{aligned}
\frac{\partial\widetilde{j}^{*}}{\partial\widetilde{L}} &= \left(\det Jb\right)^{-1}\det\begin{pmatrix}
1 & -\frac{B'_{j^{*}}B\left(\widetilde{j}^{*}\right)}{\left[1-B\left(j^{*}\right)\right]^{2}}\frac{\widetilde{L}}{L} & \frac{B\left(\widetilde{j}^{*}\right)}{1-B\left(j^{*}\right)}\\
1 & -A'_{j^{*}}\tau & 0\\
1 & 0 & 0
\end{pmatrix},\\
&= \left(\det Jb\right)^{-1}\frac{B\left(\widetilde{j}^{*}\right)}{1-B\left(j^{*}\right)}A'_{j^{*}}\tau,\\
&< 0.
\end{aligned}
$$